

Abstract

As the United States continues to rely heavily on the use of fossil fuels and other nonrenewable energy sources, the need for affordable alternative renewable energy becomes increasingly important. Solar power is an effective alternative but is not accessible to those who do not have the necessary funds to install costly panels. This project focuses on identifying optimal locations for solar farms to be constructed in Greenville County utilizing Geographic Information Systems (GIS)-based spatial analysis. By utilizing solar irradiation data, topological data, land use data, infrastructure data, and protected areas data, this study aims to map out flat, undeveloped, sun-rich areas that local communities can develop into solar farms. Despite the environmental concerns and high cost associated with solar panel materials and land development, strategically placed solar farm installations can significantly reduce nonrenewable energy use and stimulate local economic growth, aiding in Greenville's shift to renewable energy.

Background and Objectives

The United States of America continues to use a large amount of unrenewable energy. This use of fossil fuels and natural gas is further contributing to the effects of climate change seen across the globe. Renewable energy is slowly becoming more prevalent in cities and suburbs, with many individuals choosing to put solar panels on their roofs in order to cut back on energy cost. Currently, solar energy accounts for 4% of the total energy in the US, with 3.2 million homes in America having solar panels (EIA. 2021). The issue with solar panels is that they are dependent on proper conditions to function and utilize materials that require extensive mining to acquire (Al-Zoubi et al. 2024). Thus, it is very important to place solar panels strategically in order to optimize their efficiency. Many individuals choose to install these panels on their homes, but those residing in apartments, or that do not have the funds to invest in solar panels do not have this option. Solar farms, which are large plots of land that have several solar panels installed for apartments or commercial areas, provide an avenue for those who cannot install individual solar panels to have access to solar energy.

Objectives:

- Identify factors that affect solar farm land suitability
- Locate optimal locations in Greenville county for solar farms using land and environmental data
- Support the development of renewable energy infrastructure in Greenville
- Offer local residents with opportunities for investment in clean energy
- Stimulate economic growth through job creation and business for local solar development companies

Study Area & Methods

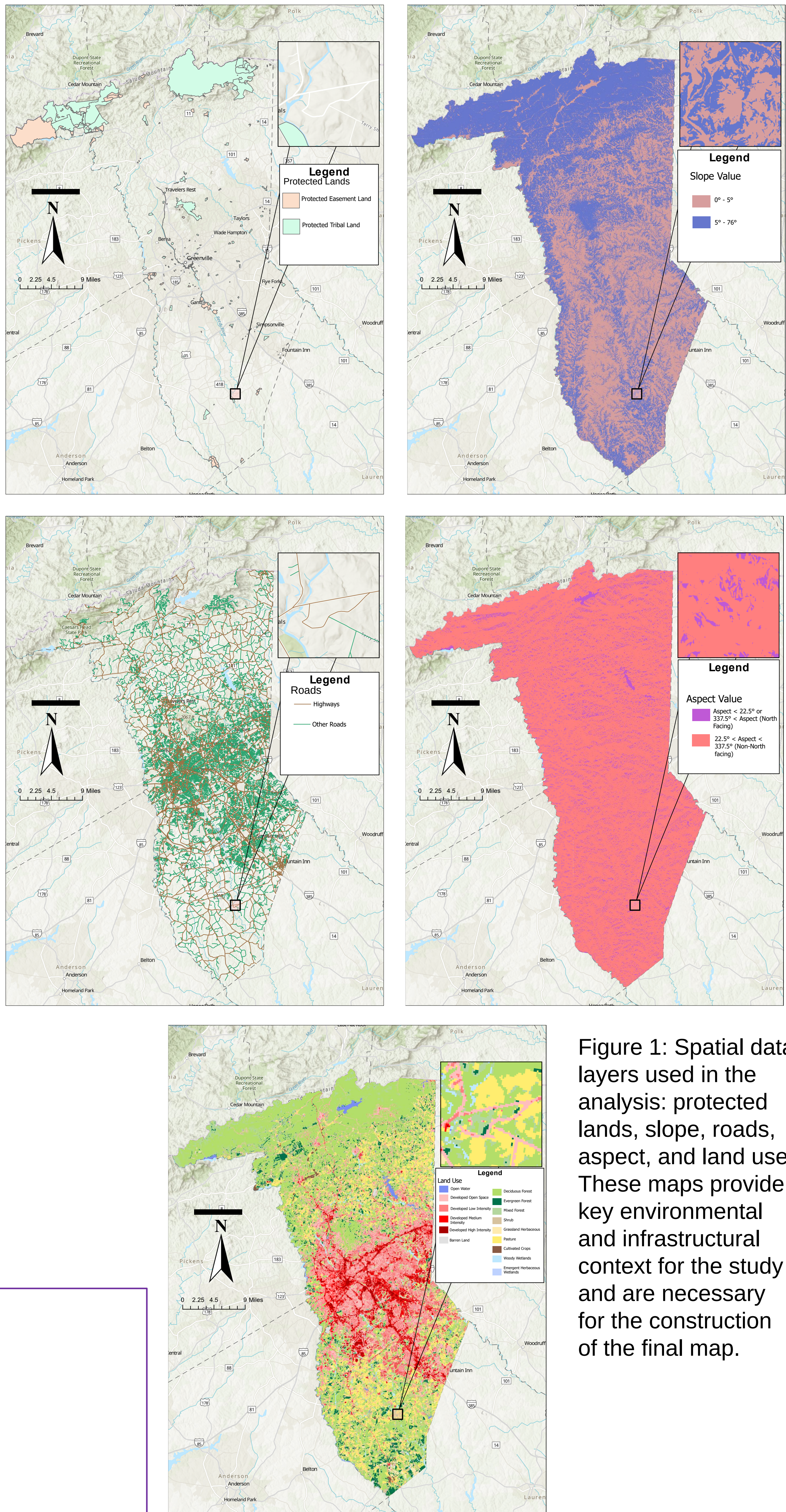
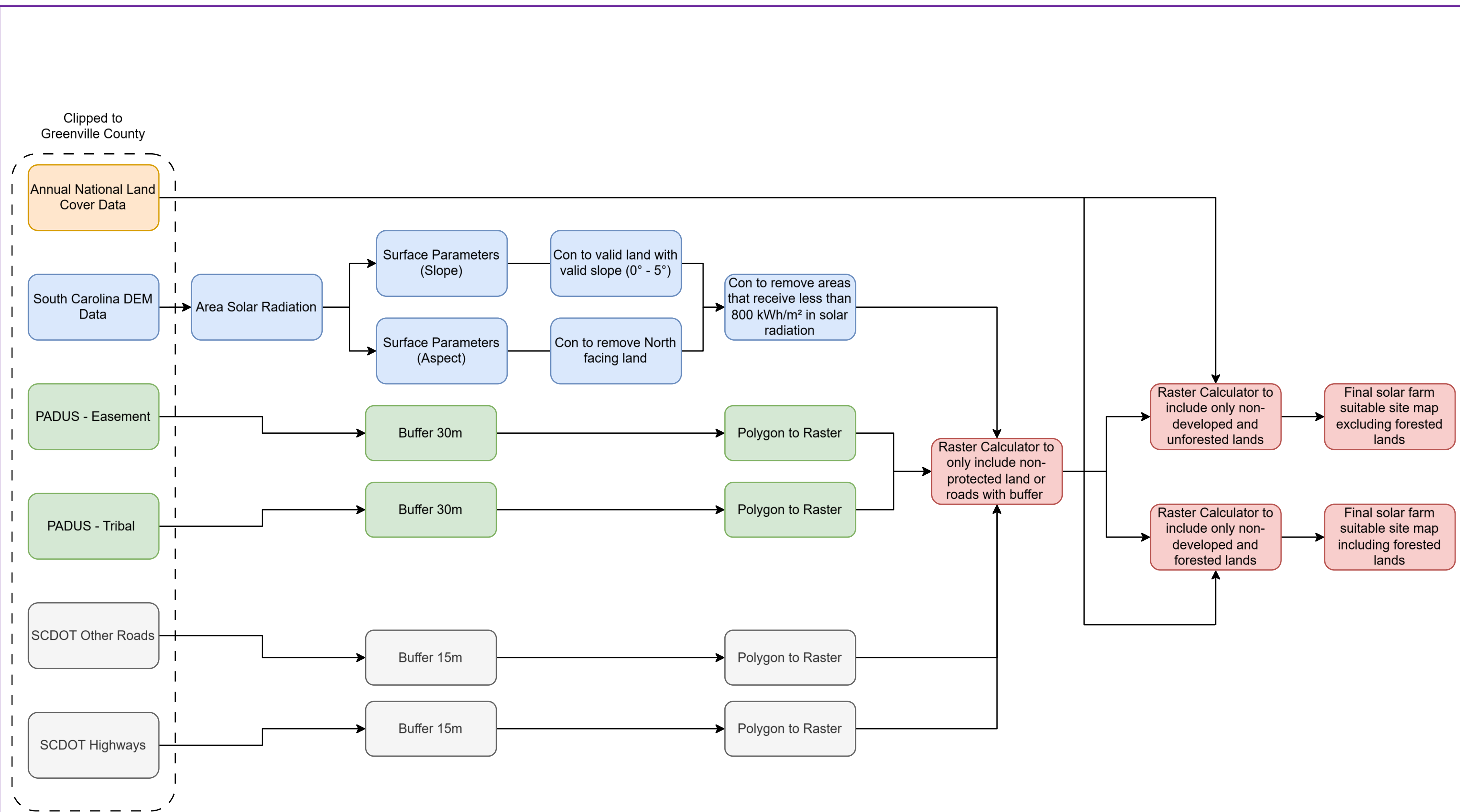


Figure 1: Spatial data layers used in the analysis: protected lands, slope, roads, aspect, and land use. These maps provide key environmental and infrastructural context for the study and are necessary for the construction of the final map.

Results

Criteria	Acceptable Range
Aspect	22.5° < Aspect < 337.5°
Slope	0° - 5°
Land Use	Barren Land, Shrub, Grassland/Herbaceous, Pasture/Hay
Solar Irradiance	At least 800 kWh/m² of solar radiation
Roads	Not within 15m of a road
Protected Lands	Not within protected land border

Figure 2: Spatial data layers and their acceptable ranges to be included in the final map. Layers with data values not found to be suitable are not included in the final map.

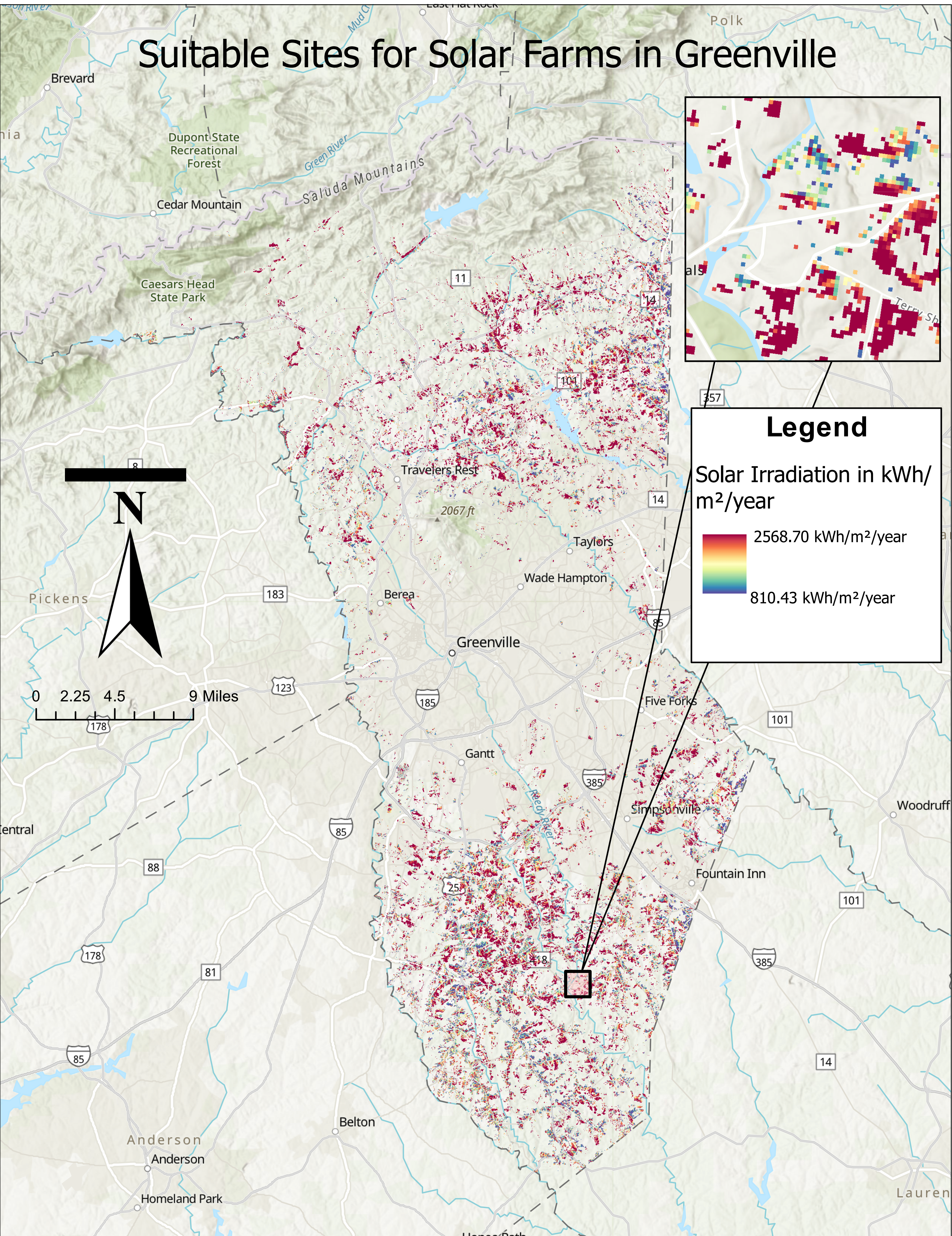


Figure 3: This map shows land that meets the criteria to be determined as suitable. Areas depicted in red/orange signify areas of high solar irradiation while areas in blue/green are areas with low solar irradiation. This map does not include suitable lands that are forested areas. Each pixel represents a 900m² area.

Conclusions

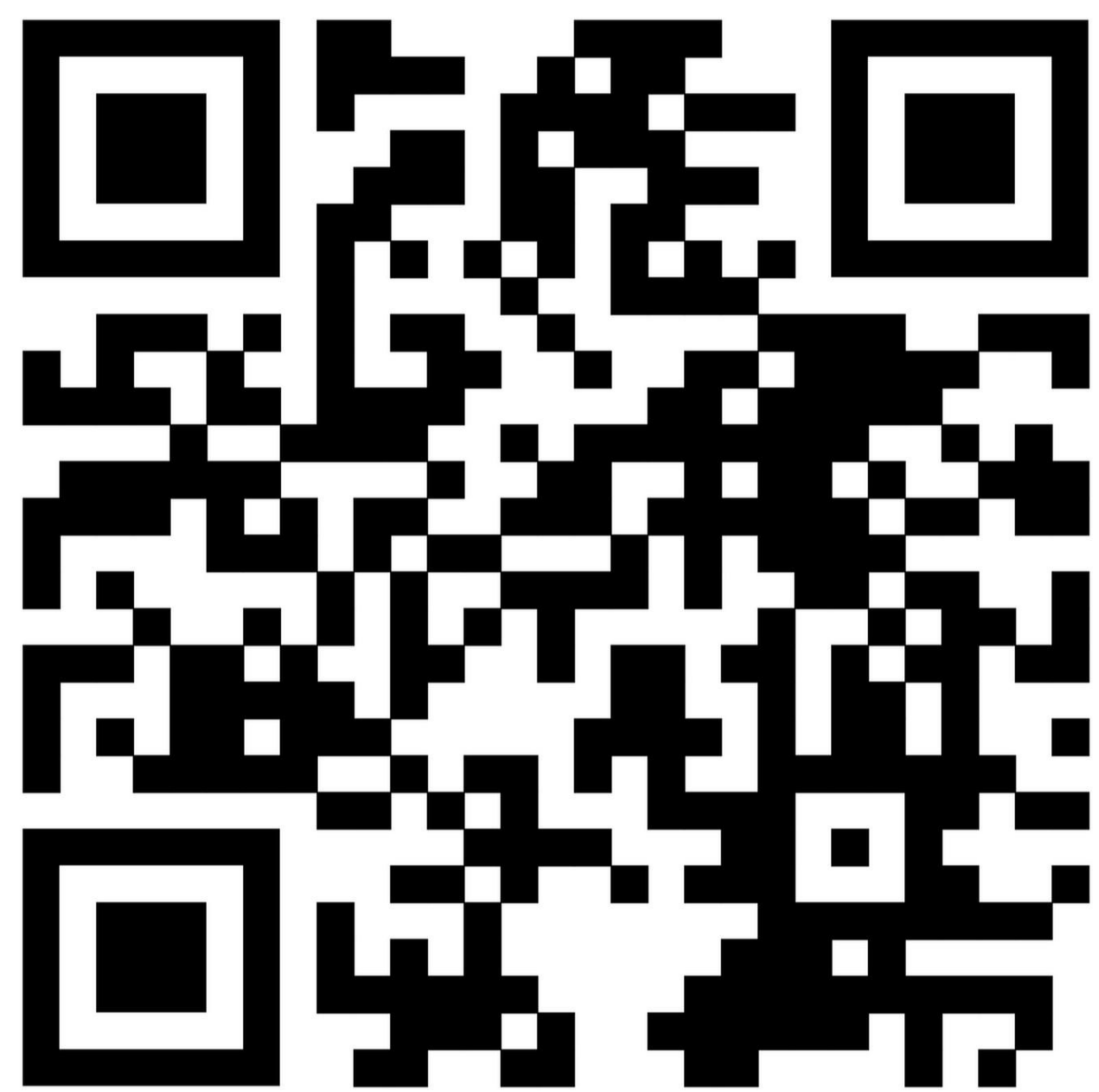
Accomplishments:

- Identified solar irradiation, slope, aspect, land use, protected areas, and adequate distance from roads as relevant factors
- GIS-based spatial analysis identifying key locations in Greenville County that have the potential for solar farm development
- Determined that most suitable land is concentrated away from downtown Greenville and residential areas

Future improvements:

- Higher resolution Digital Elevation Model for increased accuracy
- Narrowing focus to specific communities that have demonstrated interest in solar farm installation

References/Data Sources



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